LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 10 Test: The Human Eye and the Colourful World

Chapter / Topic: The Human Eye and the Colourful World

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SECTION A

Q1. The human eye can focus on objects at different distances by adjusting the focal length of the eye lens. This is due to: **[1]**

Answer: (b) accommodation

Q2. The human eye forms the image of an object at its: **[1]**

Answer: (d) retina

Q3. The least distance of distinct vision for a young adult with normal vision is about: **[1]**

Answer: (c) 25 cm

Q4. The change in focal length of the eye lens is caused by the action of the: **[1]**

Answer: (c) ciliary muscles

Q5. Myopia (near-sightedness) is corrected using a: **[1]**

Answer: (a) concave lens of suitable power

Q6. **Assertion (A):** The sky appears blue on a clear day. **Reason (R):** Blue light has a shorter wavelength and is scattered much more strongly by the fine… **[1]**

Answer: (a)

SECTION B

Q7. Why do stars twinkle? **[2]**

Answer: Starlight is refracted by the atmosphere, whose refractive index keeps changing (atmospheric refraction). Stars are very far away and act like point sources, so the apparent position of the star and the amount of light entering the eye keep changing. The star looks brighter at one moment and fainter the next.

- Atmospheric refraction with changing conditions: 1
- Flickering position/brightness: 1

OR, answer: Near the horizon, sunlight travels through a much thicker layer of air before reaching our eyes. Most of the blue light (shorter wavelength) is scattered away, so mainly red light (longer wavelength) reaches us.

- Longer path through atmosphere: 1
- Blue scattered away, red reaches eye: 1

Q8. What is dispersion of white light? What causes it? **[2]**

Answer: The splitting of white light into its seven component colours (VIBGYOR) by a prism. Different colours bend through different angles when they pass through the prism, so they separate.

- Definition: 1
- Different colours bend by different amounts: 1

SECTION C

Q9. A student can see distant objects clearly but cannot read a book held at a normal distance. **[3]**

(i) Hypermetropia (far-sightedness). [1]

(ii) The focal length of the eye lens is too long, or the eyeball has become too small. [1]

(iii) A convex lens of suitable power. It provides extra focusing power so that the image is formed on the retina. [1]

Q10. The Sun is visible about 2 minutes before actual sunrise and about 2 minutes after actual sunset. Explain why. **[3]**

Answer: This is due to atmospheric refraction. When the Sun is slightly below the horizon, its light passes from rarer to denser layers of air and bends towards the normal. So the Sun appears to be above the horizon, at a raised position. The difference between actual and apparent sunrise or sunset is about 2 minutes.

- Atmospheric refraction: 1
- Light bends, Sun appears raised: 1
- About 2 minutes earlier/later: 1

OR, answer: The Tyndall effect is the scattering of light by colloidal particles, which makes the path of a light beam visible (e.g. sunlight through a canopy or a dusty room). Danger signals are red because red light is scattered the least by fog or smoke, so it can be seen clearly from a distance.

- Tyndall effect: 1
- Red is least scattered: 1
- Visible from far: 1

SECTION D

Q11. Answer the following: **[5]**

(i) Myopia (near-sightedness): a person can see nearby objects clearly but not distant ones, because the image is formed in front of the retina. Causes: excessive curvature of the eye lens, or elongation of the eyeball. Correction: a concave lens of suitable power, which brings the image back onto the retina. [3]

- Definition (image in front of retina): 1
- Two causes: 1
- Concave lens correction: 1

(ii) With age, the power of accommodation of the eye decreases because the ciliary muscles weaken and the eye lens becomes less flexible. The near point moves farther away, making it hard to see nearby objects. It is corrected with a convex lens, or with bifocal lenses if the person also has myopia. [2]

- Cause: 1
- Correction: 1

OR, answer: (a) When white light passes through a glass prism, different colours bend through different angles and split into a band of seven colours called a spectrum (VIBGYOR). Red bends the least and violet bends the most. (b) Newton placed a second, identical prism upside down next to the first. The spectrum from the first prism recombined into white light after passing through the second.

- (a) dispersion 1, spectrum 1, red least / violet most 1
- (b) inverted second prism 1, white light recombined 1

SECTION E

Q12. A rainbow is a natural spectrum that appears in the sky after a rain shower. It is caused by the dispersion of sunlight by tiny water droplets in the… **[4]**

(i) Away from the Sun (with the Sun behind you). [1]

(ii) Tiny water droplets in the atmosphere. [1]

(iii) Refraction with dispersion, then internal reflection, then refraction again. [2]

- All three in order: 2 (two correct: 1)